Campusmonk

Number System

Class Sheet & Assignment Based on PYQ

Type I : (Unit Digit & Factors)

Question 1 : The Number of Zeros at the unit place of the product

$$81 \times 82 \times 83 \times \ldots \times 89 \text{ is}$$

TITA (type In the Answer)

Question 2:P is the product of all prime numbers from 1 to 100. Then the number of zeros at the end of the product is

(a) 0 (b) 1 (c) 24 (d) none of these

Question 3:

What is the maximum value of m if the number

N = 35x45x55x60x124x75 is divisible by 5^m ?

(a) 4 (b) 5 (c) 6 (d) 7

Concept of Unit Digit : TITA (type In the Answer)

Question 4 : The digit In unit's place of the product : $(2153)^{167}$ is - _____

Question 5 : Unit digit in $(264)^{102} + (264)^{103}$ is. _____

Question 6 : What is the unit digit of:

$$3^{1001} \times 7^{222}$$

_____

Question 7 : What will be the unit digit in the product 7^{105} ? _____

Find out the unit digit in $4^{39} \times 9^{78}$ _____

What is the unit digit of : $(2467)^{153} \times (341)^{72}$

Question 8: What is the unit digit? CAT PYQ

1! +2! +3! +4! +5! +6! +7! +8! +9!

_____

Question 8: What is the unit digit? CAT PYQ

1! +2! +3! +4! +5! +6! +7! + ---------- 99!

A) 0 B)1 C)9 D)3 E)None of these

Concept of Factors:

Question 9 :

How many divisors will be there of the number 1020?
(a) 12 (b) 20 (c) 24. (d) 36

Question 10 :

Of the Number : N= 420

1) **Find the total number of factors**
2) **Find the total number of Even factors**
3) **Find the total number of Odd factors**

Question 11:

$$N=2^5\times13^1\times17^2$$

calculate:

- Total number of factors
- Number of odd factors
- Number of even factors

Concept: Divisibility Rule

Question 12:

If a 10-digit number 5432y1749x is divisible by 72, then what is the value of (5x-4y)?

TITA (type In the Answer)

Question13:

How many pairs (A, B) are possible in the number 89765A4B

if the number is divisible by 9 and it is given that the last digit of the number is even?

Ans:

Concept: Two-Digit Number Formation & Digit Equations

Question 14

The sum of the digits of a two-digit number is 15 and the difference between the digits is 3. What is the two-digit number?

A. 69 B. 78 C. 96 D. Cannot be determined E. None of these

Question 15

The difference between two digit number and the number obtained by interchanging the positions of its digits is 36. What is the difference between the two digits of that numbers ?

a) 4 b)6 c)9 e)NOT

Question 16

In a two digit positive number, the digit in the units place is equal to the square of the digit in tens place, and the difference between the number and the number obtained by interchanging the digits is 54. What is 40% of the original number?

A.15.6 B39 C37.2. D24

Question 17

The sum of digits of a two digit number is 11. The number obtained by interchanging the digits of the given number exceeds the number by 27. Find the number.

Concept: Exponents & Divisibility

Question 18: CDS/AFCAT

$(4^{61} + 4^{62} + 4^{63})$ is divisible by

(a) 3 (b)11 (c) 13 (d) 17

Question 19:

Which one of the following will completely divide $5^{71} + 5^{72} + 5^{73}$?

(a) 150 (b) 160 (c) 155 (d) 30

Question 20:

$(2^{71} + 2^{72} + 2^{73} + 2^{74})$ is divisible by

(a) 9 (b) 10 (c) 11 (d) 13

Question 21:

In a division sum, the divisor is 3 times the quotient and 6 times the remainder. If the remainder is 2, then the dividend is:

a) 50 (b) 48 (c) 36 (d) 28

Question 22:

In a division sum, the divisor is 10 times the quotient and 5 times the remainder. If the remainder is 48, then the dividend is:

5808 5809 5484 3472

Question 23:

Consider a sequence where the nth term, tn = n/ (n + 2), n= 1, 2, ...

The value of t3 × t4 × t5 × . . .. × t53 equals: CAT 2006

A. 2/495 B.2/477 C.12/55 D.1/1485 E.1/2970

Question 24: CAT PYQ

Let a, b, c be distinct digits. Consider a two-digit number 'ab' and a three-digit number 'ccb'. If (ab)^2 = ccb and ccb > 300 then the value of b is CAT 2003

A. 0. B. 1. C. 5. D. 6. E. 7

Question 25:

How many keystrokes are needed to type numbers from 1 to 1000? GMAT PYQ

A. 2704. B. 2890. C. 2893. D. 3001. E. None of these

Question 26: CDS '23

If x and y are two distinct prime numbers, which of the following statements is ALWAYS true regarding their greatest common divisor (HCF) and least common multiple (LCM)?

HCF is x and LCM is y

HCF is 1 and LCM is x * y

HCF is x * y and LCM is 1

HCF is 2 and LCM is x + y

Question 27:

Is ba^b + b^{b+1} is even ? (If a , b are positive integers)

I. If b is even

II. If a is even

A. Statement I alone is sufficient

B. Statement II alone is sufficient

C. Both statement I and statement II together are sufficient

D. Both statement I and statement II even together are not sufficient

E. Either I alone or II alone is sufficient to answer the question .

Question 28:

If n= 1+x, where x is the product of four consecutive positive integers, then which of the following statement is/are true

1. n is odd

2. n is prime

3.n is a perfect square

A. 1 only

B. 2 only

C. 3 only

D. 1&3 only